

ANSHAJ MALHOTRA

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFILE

Electrical and Information Technology master's student in the thesis phase, with practical experience in embedded firmware, IoT, wireless communication and backend systems development. Proficient in C/C++, FreeRTOS, ESP32/nRF52, BLE, OTA, debugging, and sensor and diagnostic data in industrial IoT environments. Experienced in robust firmware/backend components, communication pipeline validation and systematic troubleshooting. Interested in secure runtime environments, ARM TrustZone/MPU-based isolation, wireless remote diagnostics for radar/measurement systems, and process automation.

Languages: German B1 (preparing for B2); English C1.

EDUCATION

Hochschule Wismar

Expected Mar 2027

Master of Engineering in Electrical and Informational Technology | Current grade: 2.3

TECHNICAL SKILLS

Programming: Embedded C/C++, C, Python, Go, Bash, SQL

Embedded & tools: FreeRTOS, Arduino, ESP32, nRF52, ARM TrustZone/MPU fundamentals, BLE, Wi-Fi, OTA, J-Link, UART, SPI, I2C, CAN, ADC, interrupts, Git CI/CD, Docker, AWS IoT Core, PostgreSQL, REST APIs

PROFESSIONAL EXPERIENCE

IoT Solution Manager Intern | DynaWo GmbH & Co. KG

2026

Ettlingen/Karlsruhe, Germany

- Built and validated wireless IoT data pipelines using Node-RED, MQTT and BLE gateways to ingest, normalize and diagnose sensor data in industrial test and integration environments.
- Supported wireless diagnostics and localization tests with BLE multi-gateway setups, RSSI data collection, timestamp handling and LiDAR-SLAM-based ground-truth validation.
- Developed Locate-IQ to evaluate and manage RTLS/IoT hardware for BLE, UWB, Wirepas, GPS and LoRaWAN; analyzed technical requirements for industrial measurement, diagnostics and integration applications.

IoT & Backend Developer | Heximpact

Jun 2023 - Sep 2023

Indore, India

- Developed FreeRTOS-based ESP32 prototypes (tasks, queues, mutexes, event groups) and BLE iBeacon features to analyze real-time behavior and device states.
- Extended IoT telemetry services, sensor/shadow handlers and AWS IoT Core/IoT Shadows for Senstra (Australia), including pagination, filtering and heartbeat adjustments.
- Used PostgreSQL, REST APIs, Postman, Git/GitHub CI/CD and automated validation to support reliable backend and device workflows in production-like environments.

Firmware Trainee | Trackonomy

Jan 2023 - May 2023

Indore, India

- Developed and optimized BLE communication modules on nRF52 SoCs (scanning, advertising, packet flow and PHY/protocol validation) and Python-based sniffer tools.
- Supported OTA update/versioning workflows, debugging and grey-box testing with 100+ test cases; contributed to structured test documentation.

PROJECT

Fuel Cell Monitoring | Embedded Firmware & Multilayer PCB

2025

- Developed a measurement and monitoring board in KiCad and embedded C/C++ firmware for ADC data acquisition, peripheral control, interrupt handling and hardware debugging.
- Designed a modular ESP32 firmware architecture for sensor data flow, configuration and validation, with relevance to robust diagnostic and measurement systems.

Technologies: C/C++, ESP32, ADC, interrupts, UART/SPI/I2C, KiCad, ESP-IDF, hardware debugging

CERTIFICATIONS

Python, AWS fundamentals, Git & CI/CD, Linux, Web Application, Arduino Robotics Workshop, and Data Structures and Algorithms (DSA).

ENGAGEMENT & INTERESTS

Block 17 student club, Hochschule Wismar: Active member for approximately 1.8 years; the only international team member among around 28 German members. Experience in teamwork, customer interaction and cultural integration.

Badminton: Winner of two tournaments; developed discipline, team spirit, resilience and adaptability to new environments.